

Topic # 9
Project Risk Management
Chapter 11 – Schwalbe

- Do software projects involve risks?
- Project plans are usually based on estimations about future. These estimations contain some degree of uncertainty. These uncertainties are the reasons of potential risks.
- Risks can adversely affect the progress of projects.
- Risk Management is the identification and evaluation of such situations, and planning necessary actions to prevent them or to reduce the impact of them.



- Project risk management is the art and science of identifying, analyzing, and responding to risk throughout the life of a project and in the best interests of meeting project objectives.
- ***Risk management*** is often overlooked in projects, but it can help improve project success by helping select good projects, determining project scope, and developing realistic estimates
- ***RISK***: "an uncertain event or condition that, if it occurs, has a positive or negative effect on a project's objectives". – PMBOK
- ***RISK***: "the chance of exposure to the adverse consequences of future events". – PRINCE2



Negative & Positive Risk

Negative Risk

- A dictionary definition of risk is "the possibility of loss or injury"
- Negative risk involves understanding potential problems that might occur in the project and how they might impede project success
- Negative risk management is like a form of insurance; it is an investment

Positive Risk

- Positive risks are risks that result in good things happening; sometimes called opportunities
- A general definition of project risk is an uncertainty that can have a negative or positive effect on meeting project objectives
- The goal of project risk management is to minimize potential negative risks while maximizing potential positive risks



Planning risk management	• Deciding how to approach and plan the risk management activities for the project
Identifying risks	• Determining which risks are likely to affect a project and documenting the characteristics of each
Performing qualitative risk analysis	• Prioritizing risks based on their probability and impact of occurrence
Performing quantitative risk analysis	• Numerically estimating the effects of risks on project objectives
Planning risk responses	• Taking steps to enhance opportunities and reduce threats to meeting project objectives
Controlling risk	• Monitoring identified and residual risks, identifying new risks, carrying out risk response plans, and evaluating the effectiveness of risk strategies throughout the life of the project

RISK STRATEGIES

Reactive	Commonly Used	Risk is handled if occurs	No preventive measures taken before hand
Proactive	Risk are anticipated before they occur	Risk management starts before technical work, risks are identified & prioritized	A risk prevention & contingency plan is prepared

Planning Risk Management

- The main output of this process is a **risk management plan**—a plan that documents the procedures for managing risk throughout a project
- The project team should review project documents and understand the organization's and the sponsor's approaches to risk
- The level of detail will vary with the needs of the project





Contingency and Fallback Plans, Contingency Reserves

- ***Contingency plans***
  - Are predefined actions that the project team will take if an identified risk event occurs
- ***Fallback plans***
  - Are developed for risks that have a high impact on meeting project objectives, and are put into effect if attempts to reduce the risk are not effective
- ***Contingency reserves or allowances***
  - Are provisions held by the project sponsor or organization to reduce the risk of cost or schedule overruns to an acceptable level; management reserves are funds held for unknown risks

IT Success Potential Scoring Sheet

Success Criterion	Relative Importance
User Involvement	19
Executive Management support	16
Clear Statement of Requirements	15
Proper Planning	11
Realistic Expectations	10
Smaller Project Milestones	9
Competent Staff	8
Ownership	6
Clear Visions and Objectives	3
Hard-Working, Focused Staff	3
Total	**100**

Broad Categories of Risk





Categories of Risk

- Project Risks
  - Risks that could prevent the achievement of the objectives given to the project manager and the project team.
  - Threatens the project plan.
  - If occur, projects may be over-schedule and/or over-budget.
  - Examples: unexpected costs, unrealistic schedules, personnel problems.
- Technical Risks
  - May affect the quality and deadline of the software being developed.
  - If occur, implementation of the software becomes difficult or even impossible.
  - Risks that are related to the analysis, design, coding, and maintenance phases.
  - For example, some hi-tech tools and languages are usually of high risk.
    - Are there any unknown bugs?
    - Is its documentation complete?
    - Will this technology be on the market in the future?

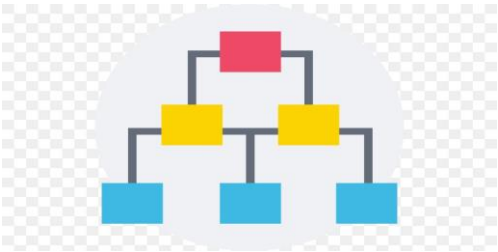
Business Risks

Developed software may not meet the requirements.

Unforeseeable Risks: natural disasters, environmental factors, regulation changes, etc.

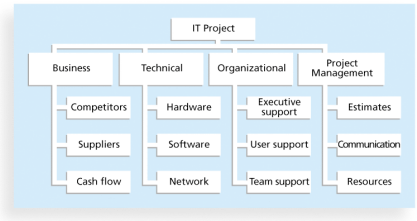
Foreseeable/anticipated Risks: Changes in the market, inflation, exchange rate variations, taxes, etc.

These risks are usually out of the direct responsibilities of the project manager and the project team.



Risk Break down Structure

- A **risk breakdown structure** is a hierarchy of potential risk categories for a project
- Similar to a work breakdown structure but used to identify and categorize risks



Potential Negative Risk Conditions Associated With Each Knowledge Area

Knowledge Area	Risk Conditions
Integration	Inadequate planning; poor resource allocation; poor integration management; lack of post-project review
Scope	Poor definition of scope or work packages; incomplete definition
Time	Errors in estimating time or resource availability; errors in determining the critical path; poor allocation and management of float; early release of competitive products
Cost	Estimating errors; inadequate productivity, cost, change, or contingency
Quality	Poor attitude toward quality; substandard design, materials, and workmanship; inadequate quality assurance program
Human resource	Poor conflict management; poor project organization and definition of responsibilities; absence of leadership
Communications	Carelessness in planning or communicating
Risk	Ignoring risk; unclear analysis of risk; poor insurance management
Procurement	Unenforceable conditions or contract clauses; adversarial relations
Stakeholders	Lack of consultation with key stakeholder



- Identifying risks is the process of understanding what potential events might hurt or enhance a particular project
- Another consideration is the likelihood of advanced discovery
- Risk identification tools and techniques include:
  - Brainstorming
  - The Delphi Technique
  - Interviewing
  - SWOT analysis

Risk Register



- The main output of the risk identification process is a list of identified risks and other information needed to begin creating a risk register
- A **risk register** is:
  A document that contains the results of various risk management processes and that is often displayed in a table or spreadsheet format
  A tool for documenting potential risk events and related information.
- **Risk events** refer to specific, uncertain events that may occur to the detriment or enhancement of the project

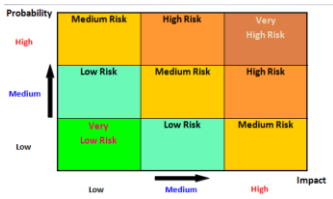
Performing Qualitative Risk Analysis

- Assess the likelihood and impact of identified risks to determine their magnitude and priority
- Risk quantification tools and techniques include:
  - Probability/impact matrixes
  - The Top Ten Risk Item Tracking
  - Expert judgment

Probability/Impact Matrix

- A probability/impact matrix or chart lists the relative probability of a risk occurring on one side of a matrix or axis on a chart and the relative impact of the risk occurring on the other
- List the risks and then label each one as high, medium, or low in terms of its probability of occurrence and its impact if it did occur
- Can also calculate risk factors:
  - Numbers that represent the overall risk of specific events based on their probability of occurring and the consequences to the project if they do occur

Risk Assessment



Risk Assessment: Probability Impact Matrix

Probability	Risk Management Approaches		
High	Significant degree of risk management is needed	Risk must be managed and controlled	Comprehensive risk management is needed
Medium	Risk can be acceptable under observation	Risk deserves risk management activities	Risk management is needed
Low	Accept the risk	Accept the risk but monitor the impact	Manage and monitor the risk
Impact	Low	Medium	High

Sample Probability/Impact Matrix

Probability	Low	Medium	High
High	risk 6	risk 9	risk 1 risk 4
Medium	risk 3 risk 7	risk 2 risk 5 risk 11	
Low		risk 8 risk 10	risk 12

Impact



- A common problem with risk identification is that a list of risks is potentially endless. A ways is needed distinguishing the damaging and likely risks. This can be done by estimating the risk exposure for each risk using formula.

risk exposure= potential damage * probability of occurrence

- Suppose a project depended on a data centre vulnerable to fire. It might be estimated that if a fire occurred a new computer configuration could be established for $500,000.
- It might also be estimated that where the computer is located there is 1 in 1000 chance of a fire actually happening, that is a probability of 0.001.
- The risk exposure in this case would be:

$500,000 * 0.001 = $500

Risk Assessment

Part of Amanda's risk exposure assessment

	Hazard	*Likelihood*	*Impact*	*Risk exposure*
R1	Changes to requirements specification during coding	1	8	8
R2	Specification takes longer than expected	3	7	21
R3	Staff sickness affecting critical path activities	5	7	35
R4	Staff sickness affecting non-critical activities	10	3	30
R5	Module coding takes longer than expected	4	5	20
R6	Module testing demonstrates errors or deficiencies in design	1	10	10



Risk Item Tracking

Top Ten Risk Item Tracking

- ***Top Ten Risk Item Tracking*** is a qualitative risk analysis tool that helps to identify risks and maintain an awareness of risks throughout the life of a project
- Establish a periodic review of the top ten project risk items
- List the current ranking, previous ranking, number of times the risk appears on the list over a period of time, and a summary of progress made in resolving the risk item

Example of Top Ten Risk Item Tracking

Risk Event	Monthly Ranking: Rank This Month	Rank Last Month	Number of Months in Top Ten	Risk Resolution Progress
Inadequate planning	1	2	4	Working on revising the entire project management plan
Poor definition	2	3	3	Holding meetings with project customer and sponsor to clarify scope
Absence of leadership	3	1	2	After previous project manager quit, assigned a new one to lead the project
Poor cost estimates	4	4	3	Revising cost estimates
Poor time estimates	5	5	3	Revising schedule estimates

Performing Quantitative Risk Analysis

- Often follows qualitative risk analysis, but both can be done together
- Large, complex projects involving leading edge technologies often require extensive quantitative risk analysis
- Main techniques include:
  - ***Sensitivity analysis***
    - (Examine how the uncertainty of the project element affects the objective in question if the other uncertain element remains the same)
  - ***Expected Monetary Value Analysis***
    - (Calculates the average outcome under uncertainty)
  - ***Decision Tree***
  - ***Simulation***
    - (Cost risk analysis is calculated using the monte carlo method)



Planning Risk Responses

- ***After identifying and quantifying risks, you must decide how to respond to them***
- Four main response strategies for negative risks



General Risk Mitigation Strategies for Technical, Cost, and Schedule Risks

Technical Risks	Cost Risks	Schedule Risks
Emphasize team support and avoid stand-alone project structure	Increase the frequency of project monitoring	Increase the frequency of project monitoring
Increase project manager authority	Use WBS and CPM	Use WBS and CPM
Improve problem handling and communication	Improve communication, project goals understanding, and team support	Select the most experienced project manager
Increase the frequency of project monitoring	Increase project manager authority	
Use WBS and CPM		

Response Strategies for Positive Risks

- Risk exploitation (Reverse of avoid)
  - **Add work or change the project to make sure the opportunity occurs**
- Risk sharing
  - **Allocate ownership of the opportunity to a third party (forming a partnership, team or joint venture)**
- Risk enhancement (reverse of mitigate)
  - **Increase the likelihood, probability and positive impacts of the risk event**
- Risk acceptance

Example

Task A needs 2 resources, each costing $100/day

Probability that the pessimistic outcome will occur is 20%

PERT Analysis
Most Likely duration is 8 days
Pessimistic duration is 20 days
Optimistic duration is 6 days
Expected duration *(20 + [8 x 4] + 6)/6*
= 9.6 = 10 DAYS

We should allow a contingency of 0.2 x $2,000 = $400

END OF TOPIC 9

- COMING UP!!!!!!
- Project HR Management
- Agile & User Stories
- Concluding topics
- FINAL EXAM
- PROJECT DEMO

